MBMA Purlin Pressure Box Test Database

Assembly of full-scale vacuum and pressure chamber purlin tests from MBMA research reports

Report by: EngineerWithData.org

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1. Introduction

This report documents how a database of full-scale cold-formed steel purlin and girt strength tests was assembled from the Metal Building Manufacturers Association (MBMA) research report library. The database holds tests in which a roof or wall assembly was loaded by air pressure in a sealed chamber, either by evacuating the chamber (vacuum) or by pressurizing it, until the purlins or girts failed. Tests loaded by concrete blocks, hydraulic jacks, point loads or axial rams are left out.

The database contains 389 strength tests from 24 MBMA reports published between 1975 and 2011, in two sets:

- **Gravity set:** 190 tests from 8 reports, plus 132 rows of supporting coupon and cross-section data.
- **Uplift set:** 199 tests from 17 reports.

MBMA 8404 contributes to both sets: its gravity tests are in the gravity set and its uplift tests in the uplift set, so no test is counted twice.

Every value in the database can be traced to a page of the source report. Measured test results are carried as printed. Calculated predictions printed in the reports are carried as legacy values, also as printed, and are labelled as predictions. Values that a report does not state are left blank rather than inferred.

2. Workflow

The database was built with a six-step pipeline. Each step is a script in the project `scripts/` folder, run as `julia --project=. scripts/<step>.jl`. Steps that read the reports call a large language model (Anthropic Claude) on page images; every step that changes a value is followed by engineering review.

Table 1: Pipeline steps

Step	What it does	Output
1. Sort	Sorts the 445 reports of the MBMA library by theme, topic and subtopic.	`01_sorted/index.csv`
1b. Select	Selects the reports that contain full-scale pressure box purlin or girt strength tests and assigns each to the gravity set, the uplift set, or both.	`02_selection/selection.csv`
2. Extract	Reads every page as an image and transcribes each data table to CSV, with column definitions, units, page numbers and notes in a comment header. Builds a standardized strength summary (Table S1) for each report.	`02_extracted/*.csv`
3. Context	Writes a brief for each report and lists the questions the reports themselves cannot answer (tributary width, dead load, units, duplicates).	`03_context/`

4. Validate	Draws random cells and claims and checks each one against the page image.	`02_extracted/validation/`
5. Harmonize	Matches equivalent columns across reports (variable concordance).	`05_sets/*_harmonized/`
6. Aggregate	Pools the reports into groups of comparable tests and audits the merge.	`05_sets/*_aggregated/`

3. Report selection

3.1. Library sort

The MBMA library holds 445 reports. The sort placed 77 of them under purlin and girt (secondary framing) research, separating full-scale tests under gravity and uplift, base-test-method reports, scale-model tests, brace-force studies and analytical work. The candidate reports were taken from the full-scale test topics and from a later sweep of the library for reports with pressure box tests that the first pass had not selected.

3.2. Selected and excluded reports

Thirty-five reports were reviewed. Twenty-four are in the database (Appendix A). Eleven were excluded for the reasons in Table 2.

Table 2: Excluded reports

Report	Reason for exclusion
MBMA 8002	Concrete-block loading.
MBMA 8201	Concrete-block loading (33 lb blocks).
MBMA 8302	Concrete-block loading.
MBMA 8802	Concrete-block loading; its 13 tests are the Rivard and Murray tests reported as data from a previous project.
MBMA 8102	Analytical study; its Table 5.1 re-lists the MBMA 7603 tests.
MBMA 8203	Observer report of the 12 tests published in MBMA 8702.
MBMA 8401	Re-publication of the nine MBMA 8310 Butler vacuum-chamber uplift tests (Appendix A, Table 2, p.29).
MBMA 8309	Edge-stiffener study on a narrow two-purlin specimen (purlins 18 in. apart under a 34 in. panel, Fig. 3); failure moments only, with no panel, fastening or bracing description.
MBMA 1205	Dissertation; Ch. 3 republishes the MBMA 1103 tests and Ch. 4 compiles other programs' tests.
MBMA 1206	Journal version of the MBMA 1103 tests.
MBMA 1207	Rotational restraint (stiffness) tests, not flexural strength tests.

3.3. Tests left out of selected reports

Some selected reports also contain tests that do not belong in the database. These rows stay in the extracted CSV files as the record of the report but are removed before the sets are pooled. Each removal is written into the header of the pooled copy.

Table 3: Tests removed from selected reports

Report	Tests left out and reason
MBMA 7603	In-plane deck shear (diaphragm) tests: load–deflection curves only, not purlin strength tests.
MBMA 8001	Beam tests B1–B7: four-point loads through channels joining the bottom flanges, no panel (p.36).
MBMA 8404	Test 13: the authors state its results are not valid because of an error in constructing the test setup (Sec. 3.3, p.34; Table 3.2 footnote). Its coupon and cross-section rows are also removed.
MBMA 8803	Test 3, the unbraced reference test first reported in MBMA 8703.
MBMA 9003	All tests except SP-10C-M: the axial-load-only tests, the combined uplift-plus-axial tests (the vacuum was held at a preset level and the specimen failed under the axial ram load, p.26), and tests whose data pages are missing from the scan.
MBMA 9410	Vacuum Box Test 1: set-up problems; no fastener failure (pp.50–51).
MBMA 9501	Proprietary manufacturer tests M1 and M2: R factor only, with no load, moment or assembly details.
MBMA 9702	Virginia Tech uplift base tests first reported in MBMA 9001 and MBMA 9102.
MBMA 9902	Axial tests A1–A4: hydraulic jack through load beams, no chamber (pp.56–57).

4. Data extraction

Each report was converted to page images and transcribed table by table. The transcription keeps every printed value as printed, including units, footnote markers and unreadable characters, which are marked `[?]`. Each CSV file opens with a comment header that gives the source, the page numbers, a definition and unit for every column, the extraction quality, and notes on anomalies.

For every report the extraction also built a standardized strength summary, Table S1. It has one row per test with the same leading columns in every report:

```
test_id, load_direction, section_type, depth_in, thickness_in, span_ft, n_spans,
n_purlin_lines, panel_type, bracing, failure_load_as_printed, failure_load_unit,
    failure_moment_as_printed, failure_moment_unit, failure_mode, value_source, pages
```

Report-specific columns, such as yield stress, predicted loads, R factors and chamber pressures, follow these. Table S1 is the primary table that the pooled groups are built from; the other tables of a report are joined to it by test name.

5. Engineering review

Every selected report was reviewed against the PDF. The review followed four rules:

1. **Measured values are used.** Where a report gives both nominal and measured dimensions or material properties, the measured values go in the main columns and the nominal values are kept in separate `_nominal` columns.
2. **Predictions are legacy values.** Predicted capacities, R factors computed by the authors and test-to-predicted ratios are carried exactly as printed.
3. **Nothing is inferred.** A value the report does not state is left blank. A value derived by hand is flagged as a hand calculation and its equation is given.

4. **Corrections are documented.** An apparent typographical error is corrected only after review, and the correction is recorded in a `# CORRECTED` note in the CSV header together with the page reference.

The context step raised 100 open questions that could not be settled from the extracted tables alone. All 100 were answered in review. The answers, with the date and the reviewer, are in `03_context/open_questions.csv`, and each answer that changed a value is also recorded as a `# REVIEW` note in the header of the affected CSV file. Table 4 lists the decisions that most affect how the data can be used; Appendix B gives the decision for every question.

Table 4: Review decisions with the largest effect on the data

Report	Decision
MBMA 1103	Measured girt thickness (Table 7) replaces the nominal footnote value. The tributary width is $L_p/2 = 6.79$ ft from Eq. 14, not the 3 ft panel width stated in the extraction. A line load in plf is added as a hand calculation from Eq. 14, which includes panel and girt self-weight.
MBMA 7501	The three continuous-span assemblies are uplift tests and are in the uplift set. The First and Second Z assemblies are distinct tests: the second was built with straightened purlins.
MBMA 8001	Measured depth and thickness (Table 3.2) replace the nominal Table 3.3 values. V1 had fixed ends.
MBMA 8404	Test 13 is removed (see Table 3).
MBMA 9003	Only SP-10C-M is kept; it includes about 9 plf of roof-system dead weight.
MBMA 9610	W_{tot} is the line load on one purlin line ($M_t = WL^2/8$ reproduces every printed M_t). No duplicates of these tests were found in any other report.
MBMA 9704	The printed failure loads include purlin self-weight (8 plf; 12 plf for FV20-3, doubled panel). FV20-4 is kept as printed although its pressure conversion uses 5.0 rather than 5.2 psf per inch of water. FV20-1 and FV20-2 had their top flanges pointing out; the other tests pointed in.
MBMA 9902	F1–F4 are kept. They were loaded by positive air pressure acting on a membrane under the panel rather than by vacuum, and the data may come from manufacturers' existing base tests (p.50).

6. Validation

The validation step drew random cells from every table and random claims from every report schema, and checked each one against the page image. A second model pass compared each extracted value with the printed value; disagreements were proposed as corrections and reviewed before they were applied.

Table 5: Validation results for the 24 reports in the database

Check type	Checked	Errors
Table cells	4,369	2
Schema claims (report summary diagrams)	208	18
Total	**4,577**	**20**

Twenty-nine corrections were proposed across all validated reports, including reports later excluded. Seventeen were applied after review, six were rejected, and six proposed on 2026-11-10 concern only the wording of the report schema diagrams and are not yet applied. None of the six pending corrections touches a value in the database. A per-table summary, with the probability that each table's accuracy

exceeds 80, 90 and 95 percent, is in `02_extracted/validation/validation_report.csv`, and each CSV header carries its own validation line.

7. Harmonization

Reports name the same quantity differently, for example `failure_load_as_printed`, `p_test_plf` or `W(tot)`. The harmonization step builds an inventory of the columns of every report in a set and clusters them into canonical variables, with the unit, the evidence and the availability in each report. The gravity concordance covers 8 reports and 65 canonical variables; the uplift concordance covers 17 reports and 68. Both are in `05_sets/*_harmonized/` together with an HTML view.

8. Aggregation

The aggregation step pools the reports of each set into groups of physically comparable tests. A reviewed plan (`05_sets/*_aggregated/_plan/plan.json`) maps each report's columns to the group variables, joins supporting tables by test name, and records constants that a report states in its notes. The plan was reviewed and edited by hand; each hand edit is recorded in the plan's notes.

8.1. Gravity set

Table 6: Gravity set groups

Group	Tests	Reports (tests)
Single-span standing seam base tests	159	9906 (118), 9002 (17), 9610 (15), 8903 (9)
Multi-span continuous lapped assemblies	24	8903 (9), 9804 (7), 9704 (6), 7506 (2)
Single-span through-fastened assemblies	7	8404 (7)
Strength tests	**190**	
Tensile coupons (supporting)	72	8404 (30), 8903 (18), 9002 (17), 9804 (7)
Measured purlin cross-sections (supporting)	30	8404 (30)
Computed AISI section properties (supporting)	30	8404 (30)

8.2. Uplift set

Table 7: Uplift set groups

Group	Tests	Reports (tests)
Simple-span through-fastened purlins and girts	116	1103 (50), 9501 (29), 8702 (12), 8310 (9), 8404 (8), 8001 (7), 9003 (1)
Standing seam base and confirming tests	40	9001 (16), 9702 (12), 9102 (8), 9902 (4)
Continuous-span through-fastened purlins	31	8703 (20), 7603 (4), 9105 (3), 7501 (3), 8803 (1)
Vacuum-box panel fastener pullover tests	12	9410 (12)
Strength tests	**199**	

8.3. Cell provenance

Every pooled CSV has a companion `_provenance.csv` with one code per cell:

- **R** (reported): copied from a table of the report.
- **C** (constant): stated once in the report's notes and applied to every test, for example a purlin spacing.
- **D** (derived): calculated by an exact definition, such as a unit conversion (kip-ft to kip-in), a ratio the report defines, or a label decoded from the report's own test-naming scheme.
- blank: not reported.

8.4. Purlin orientation

The reports describe the relative orientation of the two purlin lines in inconsistent words. The same word can mean opposite things: in MBMA 8404 "Facing" means the flanges of a C-purlin pair point toward each other (Table 1.1 sketch, p.7), whereas in MBMA 9001 "Facing" marks the tests whose flanges point the same way. The pooled column `purlin_orientation` therefore holds only "same direction" or "opposed", converted report by report from each report's own definition. The printed wording is kept in `purlin_orientation_as_printed`, and `opposed_flange_direction` gives "in" or "out" where a report states which way the flanges of an opposed pair point.

8.5. Merge audit

Each set has a merge audit (`merge_audit.csv`) that re-derives reported values from the reports' own ratios and checks unit consistency, duplicates and join coverage.

- **Gravity:** 125 identity checks, none failed. Join coverage is 92.5 percent; the remaining gap is three MBMA 9002 test labels printed inconsistently between Table S1 and Table 3.1, a table from which no columns are taken.
- **Uplift:** join coverage is 99.3 percent; the only test without a match is MBMA 8703 Test 10, which has no row in Table D and no failure load. The audit status reads "FAILED" because of two flags, both reviewed and accepted. First, the MBMA 1103 line loads do not equal pressure times tributary width, because Eq. 14 adds panel and girt self-weight. Second, the MBMA 9003 failure moment (141.8 kip-in) is flagged as a possible unit error against the much smaller MBMA 1103 girt moments; it is correct ($151.3 \times 25^2 \times 12/8 = 141.8$ kip-in).

9. Conventions and limitations

Users of the database should note the following before pooling results across reports.

1. **Dead load.** The reports treat self-weight differently. MBMA 8703 and 8803 subtract the assembly dead weight; MBMA 8404 (gravity), 9906 and 9704 include it; MBMA 7603, 8310, 8702, 9001 and 9105 report the applied chamber load only; MBMA 9804 reports the applied load and the load including self-weight in separate columns. MBMA 9704 failure loads are carried in `total_load_incl_self_weight_plf` for this reason. The group notes in each plan record the convention for each report.
2. **Load units.** MBMA 7501 and 7506 report failure loads in psf of roof area, and MBMA 7501′s are approximate text values ("about 40 psf"). These are carried in psf columns and are not converted to plf. MBMA 9501 and 9902 report moments only.

3. **Loading method.** MBMA 9902 F1–F4 were loaded by positive pressure on a membrane, not by vacuum. MBMA 1103 tests are wall girts under suction, tested the same way as the roof purlin tests.
4. **Material properties.** MBMA 9610 lists $F_y = 55$ ksi for every test of its first series, which appears to be a nominal value; the second series gives values that appear to be measured. Where a report lists several coupon values for one test (MBMA 7501), the list is carried as text.
5. **Hand calculations.** The MBMA 1103 plf line loads are a reviewer's hand calculation from the report's Eq. 14, flagged in the CSV header; the printed pressures and moments are also carried.
6. **Predictions.** Predicted loads, moments and R factors follow the specification edition used in each report and are not recomputed.

10. Data files

Table 8: Database files (project folder `Moen_purlin_testing_pull_October_2026`)

Location	Contents
`02_selection/selection.csv`	Every reviewed report with its set assignment or exclusion reason.
`02_extracted/`	Extracted tables for every selected report, with comment headers, review notes and corrections.
`02_extracted/validation/`	Validation checks, scores and proposed corrections.
`03_context/`	Report briefs, glossary, reuse caveats and the 100 answered open questions.
`05_sets/gravity_aggregated/`, `05_sets/uplift_aggregated/`	Pooled group CSVs, provenance, merge audit, variable statistics, the reviewed plan and an HTML overview (`aggregation.html`).

Appendix A: Reports in the database

Table 9: Reports in the database (G = gravity set, U = uplift set)

Report	Title	Authors (year)	Set	Tests
7501	Continuous Purlin Tests	Peköz (1975)	U	3
7506	Gravity Load Tests of C and Z Purlin Roof Systems	Linehan, Guedelhoefer (1975)	G	2
7603	Wind Uplift Tests of C and Z Purlin Roof Systems	Guedelhoefer, Boggs (1976)	U	4
8001	Ultimate Strength of Cold-Formed Steel Z-Purlins	Razak (1980)	U	7
8310	Laterally Unsupported Purlins Subjected to Uplift	LaBoube (1983)	U	9
8404	Experimental Investigation of C-Purlin Supported Roof Systems	Hendrick, Murray (1984)	G, U	7 + 8
8702	Braced Purlins Subjected to Uplift Load	LaBoube, Thompson (1987)	U	12
8703	MBMA Continuous Purlin Uplift Tests	Perry et al. (1987)	U	20
8803	Discrete Point Bracing Purlin Uplift Research	Golovin, Fisher (1988)	U	1
8903	Evaluation of the Base Test Method (gravity)	Brooks, Murray (1989)	G	18
9001	Base Test Method, Uplift Loading, Phase I	Anderson, Murray (1990)	U	16
9002	Base Test Method for Gravity Loaded Standing Seam Roof Systems	Rayburn, Murray (1990)	G	17
9003	Strength Evaluation of Strut Purlins	Hatch, Easterling, Murray (1990)	U	1
9102	Base Test Method, Uplift Loading, Phase II	Pugh, Murray (1991)	U	8
9105	Multispan C-Purlin Roof Systems Under Uplift Loading	Earls, Pugh, Murray (1991)	U	3
9410	Static Load Tests for Through-Fastened Metal Roof and Wall Systems	Ellifritt, Kreiner (1996)	U	12
9501	Uplift Capacity of Simple Span Cee and Zee Members with Through-Fastened Roof Panels	Fisher (1996)	U	29
9610	Stability of Standing Seam Roof–Purlin Systems	Fisher, Nunnery (1996)	G	15
9702	The Base Test Method for Uplift Loading	Murray (1997)	U	12
9704	Shear Plus Bending in Lapped Z-Purlins	Almoney, Murray (1998)	G	6
9804	Inflection Points as Brace Points in Multi-Span Purlin Roof Systems	Bryant, Murray (1999)	G	7
9902	Axial Strength of Purlins Attached to Standing Seam Roof Panels	Stolarczyk, Fisher (2001)	U	4
9906	Reduced Number of Gravity Loading Base Tests	Trout, Murray (2000)	G	118
1103	Flexural Strength of Exterior Metal Building Wall Assemblies with Rigid Insulation	Gao, Moen (2011)	U	50

Appendix B: Decisions on the open questions

The context step raised 100 questions that the extracted tables could not answer on their own. Each was answered by Cris Moen after checking the report; the answer below is the decision as recorded in `03_context/open_questions.csv`, with the date it was made. Questions on reports that were later excluded from the database (see Table 2) are listed too, because their answers led to the exclusions. Page numbers are PDF pages unless stated otherwise.

MBMA 7603

No.	Question	Decision
Q01 2026-10-27	What tributary width should be used to convert the chamber pressure (psf) into a line load per purlin, given that the purlins are 5 ft apart but the sealed specimen is only 6 ft 4 in. wide?	3.167 ft per purlin: 8 in. edge + half of the 5'-0" spacing, from Section 1 (Typ. @ Support Beam); 2 x 38 in. = 6'-4" sealed width. Added tributary_width_ft and failure_load_plf (psf x 3.167) to TableS1; psf kept as printed. Z 146.6/148.2, C 189.4/204.3 plf. [p4, Section 1]
Q02 2026-10-13	What were the measured yield strengths and actual (measured) thicknesses of the C and Z purlin steel and the deck?	Not measured/not reported: record measured Fy and thickness of the purlins and deck as unreported (missing), not as nominal values. The companion gravity report WJE Job No. 74565 (MBMA_7506) [p3] also gives no measured Fy or coupon data. Keep the MBMA_7603 failure loads/strengths in the dataset for later aggregation; do not drop them for lack of material properties.
Q03 2026-11-04	Is Z load stage 13 31.2 psf (as in the tabulated data, Appendix A) or 31.6 psf (as printed in the test log)?	31.2 psf; every log stage is in. water x 5.2 and Appendix A uses 31.2, so the log's 31.6 is a typo. [p73, p75]
Q04 2026-10-13	Is the C purlin centre-span thickness 0.073 in., the figure being printed as '0.773'?	0.073 in. (scan artefact of 0.773). C purlin t = 0.073 in. in the center span and 0.119 in. in the exterior spans, as printed. Correction applied and noted in MBMA_7603_Guedelhoefer1976_TableS1_StrengthSummary.csv. [p13, Fig. 4]
Q05 2026-10-13	What are the numerical peak loads and corresponding shear displacements for deck shear Tests 1, 2 and 3?	Leave out. The three deck shear tests (Fig. 7 p16, Fig. 60 p69) are in-plane diaphragm load/deflection tests on a 10 ft section cut from the already-failed Z purlin uplift specimen, reported only as curves with no numerical peak loads. Do not read peak loads from the plot; exclude the Deck shear Test 1-3 rows from the pooled uplift strength set. [p8-9, p16, p69]

MBMA 8001

No.	Question	Decision
Q01 2026-11-10	Is the beam-test failure load in lb/ft the load carried by ONE purlin (total machine load / 2 /	Not applicable: the beam tests B1-B7 are dropped (pressure box tests only). The vacuum-

No.	Question	Decision
	20 ft) or the load on the whole two-purlin assembly divided by the span?	test loads in Table 3.3 are per linear foot of purlin (p. 76).
Q02 2026-11-10	Are the depth_in and thickness_in entries in Table S1 nominal values (as in Table 3.3: 9-1/2, 8 in and .063, .110, .07 in) or the measured averages of Table 3.2 (9.45, 7.92 in and 0.060-0.115 in)?	Table 3.3 depth/thickness are nominal. Use the measured Table 3.2 averages (p. 75) in depth_in/ thickness_in; nominal values kept in depth_ nominal_in/thickness_nominal_in.
Q03 2026-11-10	For each test, do the reported ultimate stress (Table 4.1) and maximum rotation (Table 4.2) belong to the purlin that failed, or to whichever of the pair gave the larger reading?	Not stated. Tables 4.1/4.2 give one value per purlin type and are partly computed; kept as printed, not used in the strength record.
Q04 2026-11-10	Were the galvanized roof panels identical for all five purlin types, given that 'several purlin sections were made available by two manufacturers who also provided galvanized panels and their required fasteners'?	Not stated. Only one panel is shown, a 24 ga galvanized panel (Fig. 3.6); purlins, panels and fasteners came from two manufacturers (p. 28). Panel recorded as printed; supplier per purlin type not stated.

MBMA 8002 (report excluded)

No.	Question	Decision
Q01 2026-11-04	Part I is said to comprise twelve simulated live load tests, but Part II presents section properties and results for only ten (D-1 to D-3, DI-1 to DI-3, DTB-3 to DTB-6). What were DTB-1 and DTB-2 and why are they absent?	Part II covers ten tests (D-1..3, DI-1..3, DTB-3..6); DTB-1/2 are not reported here. Leave them out. [p4, p5, p14]
Q02 2026-10-27	Does the quoted uniform load per purlin include the self weight of deck, insulation, thermal bridge and purlin, or only the applied concrete blocks?	Not stated: the report does not say whether self weight (deck, insulation, thermal bridge, purlin) is included; loads carried as printed (UNIFORM LOAD PLF/PURLIN), no adjustment. [p5, p62-81]
Q03 2026-10-27	When the two purlins of a test failed at different loads (D-3: 219.5 PLF west, 198 PLF east; DI-3: 199.8 west, 189.8 east), is the Table 1 experimental value the governing purlin, the average, or something else?	Table 1 values as printed (D-3 198.0, DI-3 190.0); convention not stated, but each matches or is within 0.2 plf of the lower purlin (D-3 219.5 W/ 198 E; DI-3 199.8 W/189.8 E). DI-3 190.0 vs 189.8 flagged in the Table1 and TableS1 headers. No value changes. [p11, p62-73]
Q04 2026-10-26	In the summary table, are depth_in and thickness_in the nominal ARMCO values (8.05 in. and 0.066 in.) or measured values, and if measured, which of the four end measurements per test?	Measured values, as min-to-max ranges over the four purlin-end sheets of each test (Appendix A, pp.21-60; D4 = depth, T = thickness), no averaging. Per-end readings recorded in MBMA_ 8002_..._TableA_MeasuredPurlinDimensions.csv. ARMCO nominal 8.05 in. / 0.066 in. (p.6) kept in depth_nominal_in / thickness_nominal_in. The

No.	Question	Decision
		70 ksi on the sheets is ARMCO nominal, not measured. [p6, p7, p19, p21-60]
Q05 2026-10-26	What failure mode was observed in each roof test? Part II only alludes to lateral movement of the tension flange and gives the connector-test mode (twisting of purlin flanges, prying of the Z-connector flanges).	Not reported: Part II does not describe the observed failure mode of the roof tests (it refers to Part I, which is not in the MBMA Library); p.11 only discusses lateral movement of the tension flange as an explanation of capacity, not an observation. failure_mode left blank for the 10 roof tests; the three shear-connector tests keep 'twisting of the purlin flanges and prying of the front flanges of the Z-connector' as printed. [p6, p11, p15, p62-81]

MBMA 8201 (report excluded)

No.	Question	Decision
Q01 2026-11-04	What was the centre-to-centre spacing of the two purlins in the flexure tests?	5 ft 0 in. on centre (Addendum: flexural set-up 'purlin spacing at 5 ft. 0 in.'; Fig. 1(b)). [p17, p170-171, p174]
Q02 2026-10-26	For each test, which of the two purlins listed as 'W' and 'E' in Tables 2 and 3 is the exterior (farthest from the support joist) one?	Mapped per test from the test summary sheets ('nearer the lateral support joist' = interior; exterior = farthest, p.35): interior W for I (p48), I-A (p61), II (p73), II-A (p86), V (p139); interior E for II-B (p97), III (p114), IV (p125), VI (p150); VII not stated by side (p173), left blank. Note the question text was wrong about Test V (west is nearer). Recorded as interior_purlin / interior_purlin_page in Table 2 and the strength summary. [p35, p48, p61, p73, p86, p97, p114, p125, p139, p150, p173]
Q03 2026-10-26	Which predicted-load basis should be used when pooling, the assumed 56 ksi values of Table 4 or the measured 57.9 ksi values of Table 4A?	Report both as printed: predicted loads are calculated values, carried only as reported legacy values (Table 4, Fy = 56 ksi; Table 4A, measured Fy = 57.9 ksi, with its printed actual/predicted ratios). No basis is chosen and nothing is recomputed; the study uses the measured failure loads. No data change. [p34, p180]
Q04 2026-10-27	Should the errata (Appendix J) corrections be applied to the extracted tables, e.g. Table 4 Test II predicted load 301.7 rather than the printed 310.7, Table 3 note 56 ksi rather than 57 ksi, Table 3 column head F_c rather than F_e?	Apply. Test II predicted 310.7 -> 301.7 plf (errata; Table 3 II-W w_u 301.70) in Table4 and TableS1 predicted_load_aisi_fy56_plf. Table 3 III-W M_c 8.07 -> 8.97 ft-k (typo; computer sheet 8.972, M_u 14.98 = 1.67 x 8.97). Table 3 note 56 ksi and heading F_c recorded as header notes only. Fy 57.9 prediction and ratio columns stay as

No.	Question	Decision
		printed. Tested failure loads unchanged. [p216, p23, p73]

MBMA 8203 (report excluded)

No.	Question	Decision
Q01 2026-11-04	Is the 76 in. in the load conversion the full chamber tributary width (so 76/2 = 38 in. per purlin) or something else, and is P_u therefore strictly per purlin?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.
Q02 2026-11-04	Does P_u include the self weight of the purlin and panel (which acts opposite to uplift in the inverted setup), or is it the net chamber pressure alone?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.
Q03 2026-11-04	What roof panel profile, thickness and fastener size were used, and what rotational stiffness K was assumed in the computations?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.
Q04 2026-11-04	Were the specimens single-span simply supported (implied by 20 ft purlins and 237.5 in. span) and what were the end support/anti-roll details?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.
Q05 2026-11-04	Are the anomalous Table 2 entries (test 3 north B1 = 3.938 in., test 5 south L2 = 8.813 in., test 9 south H = 7.063 in.) typographical errors?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.
Q06 2026-11-04	Is the failure mode assignment per specimen, or only per group as described in the text?	Not used: MBMA_8203 is excluded from the pooled sets (same 12 tests as MBMA_8702); see the MBMA_8702 answers.

MBMA 8302 (report excluded)

No.	Question	Decision
Q01 2026-11-04	What was the centre-to-centre spacing of the two purlins in the full-scale flexural tests?	Not stated in this report for the flexural tests; 4 ft 10-1/2 in. applies to the diaphragm series only (p.14). Not inferred from MBMA_8201. TableS1 header line corrected (it had given 4 ft 10-1/2 in. as the flexural spacing). [p14, p16-18, p42]
Q02 2026-10-13	What numerical value of the rotational rigidity factor F was obtained from the single F-test?	Keep the F-test as companion data. F is reported only as curves, not as a single number: Fig. 8 load vs deflection (PDF p40) and Fig. 9 F (k-in/in/rad) vs deflection (PDF p41; caption misprinted as Load vs. Deflection); method p28. F rises from about 0.043 at 0.5 in. to a plateau of about 0.105 k-in/in/rad at 3.5 in. Both curves digitized into MBMA_8302_Ghazanfari1983_Fig8_9_FTestCurves.csv and linked to Tests I, III, IV, V,

No.	Question	Decision
		VI, VII (Table 1 column F) via the rotational_ restraint_F_ref column in the TableS1 strength summary; Test II (no panel-to-purlin fasteners) left blank. Panel gage of the F-test specimen not stated. [p15, p28, p38, p40, p41]
Q03 2026-10-13	What were loadings 1 (Test I) and 1-2 (Test II) that preceded the reported loadings 2 and 3?	Unknown; the report never describes Loading 1 of Test I or Loadings 1-2 of Test II. Note that Tests I-2 and II-3 may have had multiple loads applied before the reported failure loading (possibly the ~20% preload of Section 2.4 or unreported restarts; prior damage cannot be ruled out). Keep the failure loads as reported, with this note in the TableS1 strength summary header. [p4-5, p27, p51, p63]
Q04 2026-10-13	What were the numerical results of the fastener shear tests?	Omitted from this report. Fastener shear tests were conducted (Section 2.5, p29: 1.35 in. strips, two fasteners at 2 in., rupture load and failure mode recorded) and Sections 1 and 2 say results are in Section 3.9, but Section 3.9 (pp38-41) gives only coupon, F-test and diaphragm results. Record fastener shear capacity as unreported. [p13-14, p29, p38-41]

MBMA 8404

No.	Question	Decision
Q01 2026-11-04	Exactly how was the chamber suction converted into the per-purlin uniform load in plf (tributary width used, and whether all four purlins were assumed to carry equal share)?	Use the printed plf; the conversion (tributary width, sharing among purlins) is not stated. The printed values follow a constant 25.57 plf per in. of water, each interior test purlin credited the same value. [p24, p42, p52, p70]
Q02 2026-10-27	For the uplift tests, does the quoted dead load (8.9 plf) add to or oppose the applied suction in the reported failure load?	Dead load added, as printed: the assembly was turned over for uplift (p.22), so panel and purlin weight act with the suction; the report prints the sum (Test 7: 79.3 + 8.9 = 88.2 plf, p.110; Table 3.2). No change. [p22, p26-27, p110]
Q03 2026-10-27	Which of the two interior test purlins does the single tabulated reference load in Tables 3.1 and 3.2 correspond to in each test?	The tested ultimate load is the Actual Failure Load in Tables 3.1/3.2, one uniform load per purlin line, carried as printed. The single AISI predicted load and ratio are legacy predictions; which purlin's Table 2.2 w_u they correspond to is not stated (both are in Table2_2). Header note added to TableS1; no value changes. [p20, p26-27]

No.	Question	Decision
Q04 2026-10-26	Which bracing description is correct for Test 10 (Table 1.1 says no intermediate braces, Section 3.3 and Table 3.2 say braces at ends and midspan) and for Test 14 (Table 1.1 says third points, Section 3.3 says 2.5 ft each side of midspan)?	Neither source is right for both tests; the majority of the report governs. Test 10: ends and midspan bracing plus wood blocks at midspan and 2, 5, 10 ft from midspan (Sec 1.2 p8, Sec 3.3 p33, Table 3.2, B.31 p140); Table 1.1 'None' is the outlier. Test 14: ends and third points (Table 1.1 p7, Sec 1.2 p8, Table 3.2, B.71 p180); Sec 3.3 p34 '2.5 ft each side of midspan' repeats Test 13 and is a text error. Strength summary already matched; # BRACING NOTE added to Table 1.1 and the strength summary. Table 1.1 values kept as printed. [p7, p8, p33, p34, p140, p180]
Q05 2026-10-26	What gauge length was used for the coupon elongation values averaging 27.9% in Table 3.3?	Not reported: Section 2.5 (p24) says only 'standard tensile coupon tests' and Table 3.3 (p36) gives no gauge length. Elongations kept as printed with gauge length left blank; 2 in. is not assumed. Elongation does not affect the strength data. [p24, p36]

MBMA 8702

No.	Question	Decision
Q01 2026-11-04	Which purlin of the pair (north or south) governed failure, and should per-test section properties be the governing purlin's or the average of N and S?	Not stated. P_u is one chamber pressure x 38 in. tributary, so it applies equally to both purlins; pair with N and S geometry rows and flag (do not average). [8702 p17, p21; 8203 p11, p15]
Q02 2026-10-13	What numerical value of the rotational stiffness K (and the effective moment of inertia I) was used when computing P_c and P_c"?	Not recorded; K appears only as a symbol in Appendix A (pp30-33) and no rotation tests are reported. Not needed: this study uses tested values (P_u), not calculated ones. Carry P_c, P_c' and P_c" as reported legacy values only; do not recompute them or use the P_u/P_c ratios as a basis for pooling. [p21, p30-33]
Q03 2026-10-13	Are the anomalous Table 2 entries transcription errors or real measurements?	All three are typing errors (identical in both printings, pp5 and p15) and are corrected: Test 3 N B1 3.938 -> 2.938 in.; Test 5 S L2 8.813 -> 0.813 in.; Test 9 S H '9 7.063' -> 9.063 in. Corrected in MBMA_8702_LaBoube1987_Table2_PurlinCrossSectionDims.csv (and the Test 9 depth in the TableS1 strength summary), each with a CORRECTION note. [p5, p15]
Q04 2026-10-13	Is the failure mode reported per individual specimen anywhere, or only by group in the narrative?	Failure mode is uniform within each group as stated (p25): Tests 1-4 local buckling of the edge stiffener (at approx. 94 lb/ft, then unstiffened-

No.	Question	Decision
		flange behaviour to failure); Tests 5-8 local buckling of the web; Tests 9-12 (C) local buckling of the web. Assign per test by group; already so in the TableS1 strength summary failure_mode column. No per-specimen modes are reported. [p21, p22, p25]
Q05 2026-10-27	Does the reported P_u include the self-weight of the inverted purlin/panel assembly, which acts opposite to the uplift?	P_u is pressure only by the report's formula (transducer psf x 76/2/12, Table 3 note p.21); the report does not address self-weight. Carried as printed, no adjustment; specimen inverted, purlins on top (p.12). Header note added to TableS1 and Table3 test results. [p12, p17, p21]

MBMA 8802 (report excluded)

No.	Question	Decision
Q01 2026-11-04	How was the gravity load physically applied in the four new full scale tests, and was the reported plf obtained by dividing a total applied load by two purlin lines and by the span?	Loading method not described in the report; recorded as unknown. plf is per purlin line (Table 3.5 M_us = wL^2/8 per purlin). [p14, p15, p63, p64]
Q02 2026-10-14	What failure mode was observed in each of the four new full scale tests?	Not reported. No observed failure mode is described for any of the four new tests (Section 3, pp. 62-67; Appendix D, pp. 106-112); p.67 'controlled by M_neg' refers to the prediction, not an observation. failure_mode left blank. [p62-p67, p106-p112]
Q03 2026-10-14	What was the measured yield stress of the 10 in. deep Z-purlins used in the four new tests?	No measured yield stress reported. Use the assumed 55 ksi printed in Table 3.7 (footnote '*Based on assumed yield stress of 55 ksi') and Appendix D; yield_stress_ksi = 55.0 kept and marked as assumed in a # YIELD STRESS NOTE in the strength summary. 56.6 ksi (pp. 78-104) belongs to the Table 1.1 tests. Purlin dimensions/ thickness are measured (Figs. D.1-D.4). [p66, p106, p109-p112]
Q04 2026-10-14	What lap length was used at the interior support of the two-span tests, and hence where exactly along the span was M_neg evaluated?	Not reported. No lap length is given anywhere; Fig. 2.17 (p.52) and Appendix D (p.107) only state 'M_neg at beginning of lap'. Lap length left blank; M_neg carried as a printed (calculated) prediction input only, not recomputed. [p52, p66, p107]
Q05 2026-11-04	What was the purlin spacing (tributary width) in the Table 1.1 tests, and was it the same 5 ft shown for the new test setup?	5 ft for the new tests (Fig. 3.1); spacing for the Table 1.1 (Rivard & Murray 86-01) tests unspecified. [p13, p64]

MBMA 8803

No.	Question	Decision
Q01 2026-10-27	What exactly are the factors 10.4 and 5.06 in the expression 2.7(10.4)(5.06) = 142.1 lb/ft?	10.4 converts the one-leg manometer reading (in.) to chamber pressure (psf); 5.06 ft = tributary width per purlin line; the 8.5 lb/ft dead weight is subtracted, as printed. Stated in companion MBMA_8703 p.4 (same Ceco chamber) and applied on 8803 p.11. Header note added; no data change. [8803 p11; 8703 p4]
Q02 2026-10-27	Which failure-load value should be treated as authoritative for B-1: 133.6, 133.7 or 134 lb/ft?	133.6 lb/ft: the unrounded hand-calculation value (p.11) from which the report computes its ratios (0.98, 1.09). TableS1 failure_load_as_ printed changed 134 -> 133.6 (value_source hand calculation); 134 (pp.2-3) and 133.7 (p.9) recorded in a header note; Table1/Table2 keep 134 as printed. [p2-3, p9, p11]
Q03 2026-10-26	Is Se = 2.74 in^3 an effective (post-buckling-reduced) section modulus or the gross SX-T from the computer printout?	Effective (reduced) compression-fiber section modulus, not gross. Evidence: p.12 printout Ix = 14.047 in^4 is about 12% below the gross Ix (~16 in^4) of the measured section; N.A. at 4.376 in. puts the farther fiber at 5.124 in., and 14.047/ 5.124 = 2.7416 = SX-T, the smaller (compression) modulus. Compression flange flat w/t ~37 gives lambda ~0.84 > 0.673 at Fy = 55 ksi, so the section is not fully effective. Carried as printed (calculated value, not recomputed); # SECTION MODULUS NOTE added. Measured t = 0.0681 in. (p.4; coupons 0.0682-0.0684 in., p.8); base metal vs coated not stated. [p4, p8, p12]
Q04 2026-10-26	Were the Test #3 purlins from the same coil/ producer and measured by the same dimensional protocol as the B-1 purlins?	Test #3 is not new data: it is Test 3 of the MBMA Ad Hoc Task Force continuous-span uplift tests, already in the database as MBMA_8703 (Perry et al. 1987), whose values match exactly (t = .0696, Fy = 61.6 ksi, sweep -.125, 136 lb/ft) and whose Exhibit C-3 gives the per-member measured dimensions. Whether the coil/producer or measurement protocol matched B-1 is not reported. Keep originals only: Test 3 rows in the 8803 Table 1/Table 2 CSVs are dropped via DROP_ ROWS in scripts/05_harmonize.jl; the 8803 strength summary already excluded it. [p2, p3]

MBMA 9003

No.	Question	Decision
Q01 2026-11-10	Is the tabulated axial failure load the total load applied by the hydraulic ram to the pair of purlin lines, or the load per purlin line?	Not applicable: only SP-10C-M (uplift only) is kept; the combined uplift + axial tests are dropped (vacuum box strength tests only). Not stated in the report: one hydraulic ram pulls the two spreader beams joining both purlin lines (Fig. 3.1, p.24).
Q02 2026-11-10	Why do the diaphragm shear rigidity values S fed to STUDA2 differ so greatly between series (847 k for 7 in. C, 656 k for 10 in. Z, 53.2 k for 10 in. C) when all tests used 26 ga through-fastened deck, and what is in Appendix E?	Not applicable: only SP-10C-M (uplift only) is kept; the combined uplift + axial tests are dropped (vacuum box strength tests only). S is a STUDA2 input for the axial prediction; Appendix E (shear rigidity data) is missing from the scan (printed pages 54-65).
Q03 2026-11-10	Which spans apply to the three 7 in. C axial tests - the 15/17/19 ft of Table 3.2 and the data packs, or the 17/21/25 ft of Table 3.3?	Not applicable: the three 7 in. C tests are axial-load-only and dropped.
Q04 2026-11-10	For the 8 in. Z series, what value of Pa should be associated with each interaction data point given that fastener locations were never recorded?	Not applicable: only SP-10C-M (uplift only) is kept; the combined uplift + axial tests are dropped (vacuum box strength tests only). The report used Pa = 14.1 k from test SP-8Z-0 (pp.143, 150).

MBMA 9102

No.	Question	Decision
Q01 2026-10-27	Why are the failure loads in Tables 3.2 and 3.3 systematically about 5 plf (base tests) and 6-7 plf (multispan tests) higher than the same tests' loads in Table 4.1 and the Appendix data sheets?	Strength value = data sheet / Appendix E load (used in Table 4.1 and all base-test calculations); Table 3.2/3.3 values kept as printed in failure_load_table_3_2_plf. The report does not explain the constant offset (+5.00 base; +6.00 to +7.02 multispan); self-weight speculation removed from the TableS1 header. No value changes. [p25, p26, p29, p96-99]
Q02 2026-10-15	Which failure-mode classification is authoritative where the Appendix data sheets ('Local Buckle') disagree with Tables 3.2 and 3.3 ('Distortional') for UZRP/S-1, UZRP/S-3, UCRP/S-1 and UCRP/S-3?	Table 3.2/3.3 classification is authoritative (Distortional/Local defined in Sections 3.3-3.4, pp. 25-26: all rafter-braced tests Distortional, third-point-braced tests Local, UZTR/F-3 Deck). failure_mode in the strength summary now holds the Table 3.2/3.3 value; the data-sheet 'Local Buckle' labels are kept in failure_mode_data_sheet. [p25, p26, p36, p40, p50, p54]
Q03 2026-10-26	Was the UZTR/F-1 base test built with rib deck, as Table 2.1 states, or pan deck, as the base test data sheet on page 82 states?	Rib deck throughout: Table 2.1 (p.13), test ID letter R and the UZTR/F-3 sheet (p.86) give rib; the UZTR/F-1 sheet's 'Pan Type Deck with One

No.	Question	Decision
		Piece Fixed Clip' (p.82) duplicates the UZTP/F sheets (pp.66, 70) and is a copy error. panel_type corrected to 'Rib Type Deck' in the strength summary with a # CORRECTION note. [p13, p66, p70, p82, p86]
Q04 2026-10-26	What exact arithmetic converted the manometer pressure reading to the reported plf per purlin, and were the eave and ridge lines assumed to carry equal load despite the stiffening eave angle?	Not reported: the report never shows the pressure-to-plf arithmetic (data sheets say only 'Manometer used to measure load'). Use failure loads in plf as printed, with the uplift taken as equally distributed to each purlin line, since panel lengths/overhangs were designed to equalise it (p.14; spacings and overhangs p.18). No correction applied for the eave angle (p.20). [p14, p18, p20]
Q05 2026-10-27	Is the Table 4.1 entry w+_P3 = 368.8 plf for UCRP/S (O) 1991 correct, given Appendix E computes 114.30 x 100 / 42.52 = 268.81 plf?	268.81 plf (Appendix E: 114.30 x 100/42.52); Table 4.1 368.8 is a typo. Table4_1 w_pos_p3_plf changed to 268.8 with a CORRECTION note; w_P3 (negative branch, 123.0) and the ratio unchanged. [p29, p54, p97]

MBMA 9410

No.	Question	Decision
Q01 2026-10-13	How were the 'Peak Loads' in each vacuum box summary block derived, given that they do not always equal the maximum value printed in the corresponding load-cell column (e.g. Test 2 reports 442.770 lb for cell 4 while the printed column reaches 533.946 lb)?	Use the reported Peak Pressures in each summary block as the system capacity, not maxima of the printed time-history columns. Loads were logged at 0.25 s intervals (p53) but rows are printed every 2 s, and several tests had multiple attempts, so the printed rows are a coarse load history only (Test 2: summary 1.054 psi vs printed max 0.878 psi; cell #4 442.770 lb vs 533.946 lb, p100). Already the basis of failure_load_as_printed in the TableS1 strength summary; noted in its header. [p53, p99-111]
Q02 2026-10-13	How many fasteners in total (instrumented plus perimeter) were installed in each two-span vacuum box test, and at what transverse spacing?	Not stated in the report; leave the total fastener count and spacing for the two-span tests blank (unreported). Do not infer a count. The report states only 12 fasteners for the single-span system (p50) and six load cells for the two-span system (p52). [p49-52]
Q03 2026-10-13	What is the exact panel profile (rib pitch, valley width, web stiffener geometry) for the Pascoe and American sheets?	Two different profiles, one per supplier, dimensioned in Figs. 7-6 (American) and 7-7 (Pascoe), PDF p196. American: rib top half-width 0.5, web horizontal 1.5, rib height 1.125, web

No.	Question	Decision
		slant 1.875 in. Pascoe: 1, 2, 1.5, 2.5 in. Both use l_o = 6 in. Extracted to MBMA_9410_Ellifritt1996_Fig7_6_7_7_PanelRibProfile.csv. Full panel width, rib count and pitch are not stated. [p20, p73, p196]
Q04 2026-10-13	Is the fastener in the simulated and vacuum box tests the same #12 self-drilling screw used in the Standard Test, given that the vacuum pump sizing text refers to 'a single 5/8 inch diameter screw'?	#12 self-drilling screw throughout, including the simulated (component) and vacuum box (system) tests, as in the Standard Pull-Over Test (Fig. 1-2, p19). The '5/8 inch diameter screw' on p49 appears only in the vacuum pump sizing, quoting previous tests, and is not the test fastener. [p19, p49]

MBMA 9501

No.	Question	Decision
Q01 2026-10-29	Back-calculating the Table 1 tested moments from the final Appendix 2 manometer readings using psf = 5.2 x reading and a 4.27 ft tributary width reproduces most tests (e.g. 5, 7, 8, 12-14, 16-20, 26, 27) but is low by a factor of about two for Tests 1-4, 9, 10, 11, 15 and 29. Which convention applies to those tests?	Table 1 tested moments are the strength values, as printed. The report states one conversion (p.4: reading x 5.2 psf, x 4.27 ft, minus dead load) and does not explain why Tests 1-4, 9-11, 15 and 29 back-calculate to about half; no other convention is stated (one-leg explanation not adopted). Header note added to TableS1 and Table1; no value changes. [p4, p12, p19-47]
Q02 2026-11-04	What member dead load was subtracted from the applied load for each test? Only the panel dead load (1.2 psf) is stated.	Not stated: only the 1.2 psf panel dead load is given; the member dead load subtracted is not reported. Table 1 tested moments carried as printed; no member weights inferred. [p4, p19, p23]
Q03 2026-10-26	Is the printed Member Length the centre-to-centre support span used in the wL^2/8 moment calculation, or an overall member length?	Yes. L = the printed Member Length (centre-to-centre of supports). Rebuilding the Table 1 tested moments from the Appendix 2 failure readings (psf from manometer, minus 1.2 psf panel DL, x 4.27 ft tributary width, minus member self-weight) with M = wL^2/8 and L = Member Length reproduces all 29 tests within about 1%; solving for L gives 19.15-19.20 ft for the 19.17 ft tests and 29.14 ft for Test 29. span_ft values stand. [p4, p7, Appendix 2 pp. 19-47]
Q04 2026-10-26	For the proprietary M1 and M2 tests, what were the panel type, fastener spacing/position, insulation condition and failure modes?	Not reported. The report says only that the data came from two metal building manufacturers; Tables 2-10 give member type, depth, length, thickness and R, with no panel, fastener, insulation, failure mode, tested load or moment.

No.	Question	Decision
		The 64 M1/M2 rows in the Table 2-10 R-value CSVs are dropped in harmonize (DROP_ROWS); the strength summary already excludes them. [p3, pp. 8-11]
Q05 2026-10-26	Were the two purlins in each assembly nominally identical, and is the Table A1/A2 material data only for the member that failed?	Table A1/A2 data are for the failed member only: Appendix 1 gives dimensions and material 'for each of the failed members', one row per test. Whether the two purlins in each assembly were identical is not reported; not assumed. FAILED MEMBER NOTE added to Tables A1, A2 and S1. [p4, pp. 16-17]

MBMA 9610

No.	Question	Decision
Q01 2026-11-10	Is W(tot) the line load on one purlin line or the total load shared by the two purlin lines?	Per purlin line. Not stated explicitly, but M_t = W(tot) L^2/8 reproduces every printed M_t and M_t is compared with the single-purlin M_n (Tables 1-2, p.4).
Q02 2026-11-10	What purlin spacing / tributary width relates the failure load in psf (Tables 4, 5) to W(tot) in lb/ft (Table 1)? The implied width varies from about 4.1 to 4.8 ft across the eight tests.	Not stated. W(tot)/pressure varies by test (e.g. 93-1: 4.41 ft); psf and lb/ft are carried separately and no tributary width is back-calculated.
Q03 2026-11-10	Did the opposed-purlin tests (94-1 to 94-7) include the edge angle and the 6 in. end clip, and was the diaphragm anchored?	Not stated. Edge angle and clip are described only for 93-5 to 93-8, diaphragm unanchored (p.4); bracing left blank for the 94 tests.
Q04 2026-11-10	Is the 55 ksi in Table 1 a nominal grade or a measured coupon value?	Not stated. Carried as printed: 55 ksi for all 93 tests looks nominal; 94 values (54.89-57.33) appear measured (p.4).
Q05 2026-11-10	What standing seam panel profile and clip were used, and were they the same for the 93- and 94-series?	Not stated. panel_type = standing seam; clip identification left blank (only 'SSR Clip Tab' failures and one fastener per panel flat, p.4).

MBMA 9702

No.	Question	Decision
Q01 2026-10-29	Why do the Table 1 CECO base-test loads (39.4, 66.6, 128.1, 162.5, 86.3, 152.7 plf) differ from the Appendix 'FAILURE LOAD' values (39.87, 69.91, 133.03, 162.38, 90.12, 158.27 plf), and likewise 3S5 (129.2 vs 126.9)?	No reason given; differences are not a constant offset (-5.57 to +2.3 plf). CECO strength values = test-sheet FAILURE LOAD (39.87, 69.91, 133.03, 162.38, 90.12, 158.27; confirming 64.56, 106.96, 178.52, 220.35, 126.9, 235.9) in failure_load_as_printed; Table 1 kept in failure_load_table_1_plf (renamed from appendix_failure_load_plf). Predictions/ratios legacy from Table 1. No journal/conference publication of the CECO

No.	Question	Decision
		tests found to confirm (searched 2026-10-29). [p3, p7-8, p12-34]
Q02 2026-11-04	Exactly how was the plf value computed from the chamber pressure - per purlin line at the full purlin spacing, or over a different tributary width (the point of the acknowledged tributary-area error)?	Not stated for the CECO tests beyond 'Opposed Purlins; Spacing = 5 ft.'; treat plf as per purlin line and do not back-convert to psf. [p3, p10]
Q03 2026-10-26	What were the span lengths, purlin spacings, depths, thicknesses and yield stresses for the nine Virginia Tech test sets?	Not reported here: p.6 says data for the Virginia Tech tests are in the previously cited research reports. Left blank. The VT rows are dropped in harmonize (first reported in MBMA_9001/ MBMA_9102, whose rows carry the geometry). Table 1 CSV note that decoded the VT designation letters reworded, since this report does not define them. [p6, p8]
Q04 2026-10-26	For each set, did Equation 3a (negative moment, M_us) or Equation 3b (positive moment, M_ AISI) control the predicted load?	Not reported: Table 1 gives only the predicted confirming load; p.6 says the prediction is the minimum of Eqs. 3a and 3b without saying which governed. Predicted loads carried as legacy values; controlling equation not recomputed. [p5, p6, p8]
Q05 2026-10-26	Which standing seam panel profiles and clip types (fixed or sliding) were used in each set? The VT codes P and R and the CECO tests are not linked to named products.	Not reported: CECO sheets say only standing seam; p.6 says nominally identical purlins, panels, clips and bracing were used within each base/confirming set. CECO panel and clip left unspecified; VT panel/clip types come from MBMA_9001/MBMA_9102. [p6, pp. 12-34]
Q06 2026-10-26	MBMA_9702 states the Virginia Tech uplift failure loads in MBMA_9001/MBMA_9102 were reported with a tributary-area error and prints corrected values in Table 1; which loads go into the dataset?	Keep the original measured loads as the strength values (keep-originals rule). Added failure_load_ corrected_9702_plf and a CORRECTION NOTE to the MBMA_9001 and MBMA_9102 strength summaries, filled for 7 of 9 VT sets where the match is clear; left blank for the two U-Z-T-P/F (O) sets (ambiguous match) and tests not in 9702 Table 1. VT rows in MBMA_9702 stay dropped. [p3, p7, p8]

MBMA 9704

No.	Question	Decision
Q01 2026-11-10	Why is the pressure-to-line-load factor 3.75 ft used for every test when purlin spacing was 4 ft 11 in. in FV20-1/-2 and 5 ft 0 in. in the other four tests?	Not stated. 3.75 ft is the report's own factor for every test; printed line loads carried as they are.

No.	Question	Decision
Q02 2026-11-10	Is the FV20-4 pressure conversion (16.5 in. water -> 82.5 psf, a factor of 5.0) an error, given every other test uses 5.2 psf per inch?	Keep 317.4 plf as printed (agrees with Table 3.2); flag the 5.0 vs 5.2 factor inconsistency (p.55). No correction.
Q03 2026-11-10	The predicted failure loads in Table 3.2 are single values, but each test has separate north and south purlin line geometry and separate RSD runs - which line governs the tabulated prediction?	Not stated (p.14). Predicted load carried as printed, as a single value for the system.
Q04 2026-11-10	Were FV20-1 and FV20-2 opposed (flanges in) as Chapter 2 states, or flanges out as their appendix summary sheets state?	Follow the per-test summary sheets: FV20-1/-2 top flanges point out (pp.22, 31), the others point in; conflict with Chapter 2 (p.7) noted. purlin_ orientation column added.
Q05 2026-11-10	Are 'FV20-3' (summary sheet dated 24 November 1997) and 'FV20-3A' (plots and Figure 3.1 dated 24 September 1997) the same test?	Same test. Text uses FV20-3A data for FV20-3 (p.15); sheet months appear clerical (FV20-4 photos 9/26/97 vs Nov 26 sheet). 339.6 plf kept.
Q06 2026-11-10	Why is the self-weight allowance 12 plf for FV20-3 (printed with a minus sign but added) when all other tests use 8 plf?	12 plf is correct (doubled panel, p.42) and is added, as the printed total 339.6 shows. No change.

MBMA 9804

No.	Question	Decision
Q01 2026-11-04	What exact tributary width (or pressure-to-line-load calibration) was used to convert chamber pressure to the applied line load? The printed data imply 19.50 plf per inch of water at Virginia Tech and 20.15 plf per inch at Ceco, i.e. about 3.75 ft and 3.88 ft, which match neither the 5 ft purlin spacing nor the 3.5 ft half-panel width.	Not stated. The printed loads follow constant calibrations of 19.50 plf per in. of water (Virginia Tech) and 20.15 plf per in. (Ceco), on every summary sheet; no tributary width derived. Use the printed plf. TableS1 header line corrected (it had stated a 5 ft tributary width). [p26-30, p78, p93, p107]
Q02 2026-11-04	Does the 'Experimental Failure Load' column of Tables 5.2-5.4 include self weight? The text says it does, but Test 1 is listed as 320.8 plf (the applied load, whose total is 329.82) while Test 2 is listed as 142.4 plf (neither the applied 141.7 nor the total 150.6).	Despite the p.65 note, Tables 5.2-5.4 list the applied load, not the total (Tests 1, 3, I.P. 1, I.P. 3 equal Table 3.2; Test 2 142.4 vs 141.7; Test 4 280.3 vs 280.1). TableS1 already carries applied (failure_ load_as_printed) and total (total_load_incl_self_ weight_plf). Header note added to Table5_2/5_3/ 5_4 and TableS1; ratios legacy as printed; no value changes. [p38, p65, p64-67, p78, p93]
Q03 2026-11-04	Which applied failure load for I.P. Test 2 Z-SS is correct, 100.8 plf (Table 3.2) or 102.8 plf (test summary and Tables 5.2-5.4)?	102.8 plf: test sheet p.150 (110.82 total - 4.00 deck - 4.02 purlin = 102.80; 5.10 in. x 20.15 plf/in.), as in Tables 5.2-5.4. TableS1 changed 100.8 -> 102.8 with a CORRECTION note; Table3_2 keeps 100.8 as printed (corresponds to 5.0 in.). [p38, p150]
Q04 2026-10-26	Why do the yield stresses used in the strength calculations (Table 5.1: 50.6, 87.5, 75.0 ksi) differ	Not reported. Table 3.1 = average of the coupons' 0.2% offset yields (p.37). Table 5.1 uses a single

No.	Question	Decision
	from the reported coupon averages (Table 3.1: 50.0, 87.7, 75.2 ksi)?	coupon for Test 2 (50.6, CH 14, p.100) and Test 4 (75.0, p.129), and for Test 3 a value (87.5) that matches no coupon's 0.2% offset (it equals the p.113 coupon's 0.1% offset). Strength summary keeps Table 3.1 as measured Fy; Table 5.1 kept as printed as legacy prediction inputs, not recomputed. YIELD NOTE added to Table 5.1 and the strength summary. [p37, p64, pp. 99-100, 113-114, 129-130]
Q05 2026-10-26	Is the 2.000 in. gage length printed for the Test 1 coupon correct, given that every other coupon sheet lists about 8.0 in.?	2 in. gage length for every coupon (Cris Moen). Test 1 sheet prints 2.000 in. (p.84); the other sheets print about 8.0 in. (7.991-8.015), longer than their printed total length of 8.0 in., taken as an entry error. Elongations are comparable across tests; values unchanged. GAGE LENGTH NOTE added to Table 3.1. [p84, pp. 99-156]

MBMA 9902

No.	Question	Decision
Q01 2026-11-10	Is the 'failure load' in Table 1 the total jack load on the two-purlin assembly or the load per purlin?	Not applicable: axial tests A1-A4 are dropped (pressure box tests only).
Q02 2026-11-10	How was the 'effective section modulus' in Table 5 computed — at what stress level, and with which effective width assumptions?	Not stated beyond CFS v3.04 (p.37). S_e and the bending stress M/S_e are computed values, carried as printed.
Q03 2026-11-10	What was the panel clip spacing along the purlin in each of the four test assemblies?	Equal to the panel coverage: 24 in. F1, 16 in. F2, 18 in. F3/F4 (p.51); one clip per panel edge (p.53). Recorded in a note only.
Q04 2026-11-10	How many purlin lines were present in each axial and each flexural test assembly?	Two purlin lines (east/west purlin, p.180; p.58). No change.
Q05 2026-11-10	Were the pre-existing uplift base tests single-span or continuous-span?	Single span: n_spans = 1 (pp.51, 58, 59, 60; AISI Base Test Method). F1-F4 kept as pressure-chamber (positive pressure) tests.

MBMA 9906

No.	Question	Decision
Q01 2026-11-04	What tributary width was actually used to turn the chamber pressure into the per-purlin applied line loading?	Not stated. The printed loads follow about 19.5 plf per in. of water (same Virginia Tech constant as MBMA_9804); S2 Test 10 deviates (19.07). No tributary width derived; use the printed plf. [p18, p20, p54-80]
Q02 2026-11-04	Do the six Appendix A tests with 30 ft spans, 6 in. insulation and a 'Ridge Purlin Correction'	Keep them in Series 1 (Table 4.1's count of 11 includes them; Fig. 4.1 plots them) but tag as 30

No.	Question	Decision
	really belong to Series 1, given that Table 4.1 lists Series 1 as 25'-0" span with no insulation?	ft span, 6 in. insulation, C3 clips, ridge correction included; Table 4.1's '25 ft, no insulation' is wrong for these six. [p17, p22-23, p28, p53-64]
Q03 2026-10-26	Is per-test measured flange width available for every test, or only the nominal 2 1/2 / 3 1/2 in. from Table 4.1?	Measured for 61 of 118 tests: per-test 'Measured Dimensions' drawings give top and bottom flange widths of the ridge and eave purlins for Series 1 Tests 01-04 and 17, Series 2 Tests 05-06, Series 7-8 (Tests 63-90), Series 9 (91-102) and Series 10 (105-118). None for Series 1 Tests 1-6 (Z10)-C3, the rest of Series 2, Series 3, 5 and 6 (both 3-1/2 in. series) or Tests 103-104. Transcribed to MBMA_9906_Trout2000_TableA_MeasuredFlangeWidths.csv; Table 4.1 nominal width added to the strength summary as flange_width_nominal_in. [p22, pp. 185-614]
Q04 2026-10-26	Are the 'floating' clips of Series 8-10 physically the same hardware as the 'sliding' clips of Series 1-3, or only a change in wording?	Wording only: p.22 defines C1 = Low Sliding, C2 = Low Fixed, C3 = Tall Sliding for all series; every series description including 8-10 says 'two separate sliding clip heights, and one fixed clip height' (pp. 23-27); p.25 calls Series 5's tall sliding clip 'the tall floating clip'. Floating = sliding, high = tall. Key clip type on the -C1/-C2/-C3 suffix. Not reported whether the physical clip was the same across the pan-panel (1-3) and rib-panel (5-10) series; not assumed. [p22, pp. 23-27]

MBMA 1103

No.	Question	Decision
Q01 2026-11-05	Is the R-factor column in Appendix C (and Table S1) the raw test value or the value normalised to C = B/2?	Raw, as printed: R = M_t/(S_e F_y) at the measured C (Eq. 16, p. 29); the normalisation to C = B/2 (Sec. 4.9.2, p. 30) appears only in Figures 40-43. Carried as printed.
Q02 2026-11-05	Table 4 (insulated rotational restraint test matrix) and Table 8 (average screw location C per test) are printed as empty grids; what were their contents?	Both tables are printed in the report; the page transcription missed them. Table 8 C values equal the Appendix C C column. Table 4 is the rotational-restraint (stiffness) test matrix, outside the strength database; not extracted.
Q03 2026-11-05	Each test had two parallel girts but generally only one failed; are the reported cross-section dimensions, F_y and C those of the failed girt only, and was the pressure shared equally between the two girts?	S_e, F_y, C and t are for the failed girt (p. 21). Equal load sharing is the report's own assumption (Eq. 14: each girt carries L_p/2 = 6.79 ft). Carried as printed with a note.

No.	Question	Decision
Q04 2026-11-05	Because the assemblies were tested upside down, does the panel and girt self weight add to or oppose the suction-induced moment?	Self weight adds to the suction, as written in Eq. 14 (p. 21); the assemblies were tested upside down (p. 17).